7.1 Suppose that we decompose the schema $R = (A, B, C, D, E)$ into

$$R_1 = (A, B, C)$$
$$R_2 = (A, D, E).$$

Show that this decomposition is a lossless decomposition if the following set F of functional dependencies holds:

$$A \rightarrow BC$$
$$CD \rightarrow E$$
$$B \rightarrow D$$
$$E \rightarrow A$$

A decomposition $\{R_1, R_2\}$ is loseless decomposition if

$R_1 \cap R_2 \rightarrow R_1$ or $R_1 \cap R_2 \rightarrow R_2$

since A is in the both of them so $R_1 \cap R_2 = A$

7.2 List all nontrivial functional dependencies satisfied by the relation of Figure 7.18.

A	B	C
a_1	b_1	c_1
a_1	b_1	c_2
a_2	b_1	c_1
a_2	b_1	c_3

Figure 7.18 Relation of Exercise 7.2.

The nontrivial functional dependecies $A \rightarrow B$ and $C \rightarrow B$ and dependency they logically imply: $AC \rightarrow B$

There are 19 trivial funtional dependencies of the form $\alpha \rightarrow \beta$, where $\beta \subseteq \alpha$

7.3 Explain how functional dependencies can be used to indicate the following:

- A one-to-one relationship set exists between entity sets *student* and *instructor*.
- A many-to-one relationship set exists between entity sets *student* and *instructor*.

1- The functional dependcies p(student) Primary key of student → p(instructor) and p(instructor) → p(student) indicate one-to-one relationship because any two tuples with the same value for student must have the same value for instructor, and any two tuples agreeing on instructor must have the same value for student.

2- The functional dependency p(student) → P(instructor) indicates a many to one relationship since any student value which is repeated will have the same instructor value, but many student values may have the same instructor value.

7.4 Use Armstrong's axioms to prove the soundness of the union rule. (*Hint*: Use the augmentation rule to show that, if $\alpha \rightarrow \beta$, then $\alpha \rightarrow \alpha\beta$. Apply the augmentation rule again, using $\alpha \rightarrow \gamma$, and then apply the transitivity rule.)

To prove that:

if $\alpha \rightarrow \beta$ and $\alpha \rightarrow \gamma$ then $\alpha \rightarrow \beta\gamma$

following the hint, we derive:-

$\alpha \rightarrow \beta$ given $\alpha\alpha \longrightarrow \alpha\beta$ augmentation rule $\alpha \rightarrow \alpha\beta$ union of identical sets $\alpha \rightarrow \gamma$ given $\alpha\beta \rightarrow \gamma\beta$ augmentation rule $\alpha \rightarrow \beta\gamma$ transitivity rule and set union commutativity

7.5 Use Armstrong's axioms to prove the soundness of the pseudotransitivity rule.

Proof using Armstrong's axioms of the pseudotransitivity rule:

if $\alpha \rightarrow \beta$ and $\gamma\beta \rightarrow \sigma$ then $\alpha\gamma \rightarrow \delta$

proof:

$\alpha \rightarrow \beta$ given $\alpha\gamma \rightarrow \gamma\beta$ augmentation rule and set union commutativity $\gamma\beta \rightarrow \delta$ given $\alpha\gamma \rightarrow \delta$ transitivity rule

7.6 Compute the closure of the following set F of functional dependencies for relation schema $R = (A, B, C, D, E)$.

$$A \rightarrow BC$$
$$CD \rightarrow E$$
$$B \rightarrow D$$
$$E \rightarrow A$$

List the candidate keys for R.

Starting with $A \rightarrow BC$, we can conclude that $A \rightarrow B$ and $A \rightarrow C$

Since $A \rightarrow B$ and $B \rightarrow D$ then $A \rightarrow D$ (decomposition, transitive)

Since $A \rightarrow CD$ and $CD \rightarrow E$ then $A \rightarrow E$ (union, decomposition, transitive)

Since $A \rightarrow A$, we have $A \rightarrow ABCDE$ from the above steps

and since $E \rightarrow A$ so $E \rightarrow ABCDE$

since $CD \rightarrow E$ so $CD \rightarrow ABCDE$

since $B \rightarrow D$ and $BC \rightarrow CD$ then $BC \rightarrow ABCDE$

The candidate keys are A, E, CD, BC

7.29 Show that the following decomposition of the schema R of Exercise 7.1 is not a lossless decomposition:

$$(A, B, C)$$
$$(C, D, E).$$

Hint: Give an example of a relation $r(R)$ such that $\Pi_{A,B,C}(r) \bowtie \Pi_{C,D,E}(r) \neq r$

A	B	C	D	E
1	6	5	7	3
2	8	5	9	4

A	B	C
1	6	5
2	8	5

C	D	E
5	7	3
5	9	4

A	B	C	D	E
1	6	5	7	3
1	6	5	9	4
2	8	5	7	3
2	8	5	9	4

→ the decomposition is a lossy decomposition

7.30 Consider the following set F of functional dependencies on the relation schema (A, B, C, D, E, G):

$$A \rightarrow BCD$$
$$BC \rightarrow DE$$
$$B \rightarrow D$$
$$D \rightarrow A$$

a. Compute B^+.

b. Prove (using Armstrong's axioms) that AG is a superkey.

c. Compute a canonical cover for this set of functional dependencies F; give each step of your derivation with an explanation.

d. Give a 3NF decomposition of the given schema based on a canonical cover.

e. Give a BCNF decomposition of the given schema using the original set F of functional dependencies.

a. $B^+ = A, B, C, D, E$

b. $A \rightarrow BCD$, $A \rightarrow BC$

$BC \rightarrow DE$

$A \rightarrow DE$ (transtivity)

Thus, $A \rightarrow BCDE$

By augmentation rule $AG \rightarrow ABCDEG$

$\therefore$ proves that AG is a superkey

c. D is extraneous in $A \rightarrow BCD$ so, remove it

D is also extraneous in $BC \rightarrow DE$ so remove it

$\therefore A \rightarrow BCBC \rightarrow EB \rightarrow DD \rightarrow A$

d. A, B, C, B, C, E, B, D, D, A, A, G

e. A, B, C, B, D, A, E, A, G

7.31 Consider the schema $R = (A, B, C, D, E, G)$ and the set F of functional dependencies:

$$AB \rightarrow CD$$
$$B \rightarrow D$$
$$DE \rightarrow B$$
$$DEG \rightarrow AB$$
$$AC \rightarrow DE$$

R is not in BCNF for many reasons, one of which arises from the functional dependency $AB \rightarrow CD$. Explain why $AB \rightarrow CD$ shows that R is not in BCNF and then use the BCNF decomposition algorithm starting with $AB \rightarrow CD$ to generate a BCNF decomposition of R. Once that is done, determine whether your result is or is not dependency preserving, and explain your reasoning.

AB → CD is not a trivial functional dependency
Also, $(AB)^+$ = A, B, C, D, E that is AB is not superkey
Thus the relation R is not in BCNF

BCNF decomposition of R:-

A, B, G, A, B, E, A, B, C, B, D

Clearly the above decomposition isn't dependency preserving since the functional dependency DE → B cannot be tested by using only one relation

7.32 Consider the schema $R = (A, B, C, D, E, G)$ and the set F of functional dependencies:

$$A \rightarrow BC$$
$$BD \rightarrow E$$
$$CD \rightarrow AB$$

a. Find a nontrivial functional dependency containing no extraneous attributes that is logically implied by the above three dependencies and explain how you found it.

b. Use the BCNF decomposition algorithm to find a BCNF decomposition of R. Start with $A \rightarrow BC$. Explain your steps.

c. For your decomposition, state whether it is lossless and explain why.

d. For your decomposition, state whether it is dependency preserving and explain why.

c. it's lossless decomposition

d. our decomposition isn't dependency preserving. the functional dependency $BD \rightarrow E$ isn't preserved.

* from web this answer

a. I claim that such a functional dependency doesn't exist. Suppose the countrary that such functional dependency exists, say $\alpha \rightarrow \beta$ Thus $\alpha \rightarrow \beta$ is a nontrivial functional dependency containing no extraneous attributes and it's logically implied by F. Define $F_1 := F \cup \alpha \rightarrow \beta$. Now consider an attribute $x \in \beta$, I'm going to prove that x is extraneous.
Consider the set $\Rightarrow F_1' = (F_1 - \alpha \rightarrow \beta) \cup \alpha \rightarrow (\beta - x)$
since the functional dependency $\alpha \Rightarrow x$ can be inferred from F_1', x is extraneous in β, $\therefore$ such a functional dependency doesn't exist.

b. by using the functional dependency $A \rightarrow BC$ we decompose the schema into A, D, E, G, A, B, C note that A, B, C is in BCNF. By applying Augmentation followed by transitivity rule $AD \rightarrow E$ $\therefore$ A, D, G, A, D, E, A, B, C

7.33 Consider the schema $R = (A, B, C, D, E, G)$ and the set F of functional dependencies:

$$AB \rightarrow CD$$
$$ADE \rightarrow GDE$$
$$B \rightarrow GC$$
$$G \rightarrow DE$$

Use the 3NF decomposition algorithm to generate a 3NF decomposition of R, and show your work. This means:

a. A list of all candidate keys

b. A canonical cover for F, along with an explanation of the steps you took to generate it

c. The remaining steps of the algorithm, with explanation

d. The final decomposition

> a. A list of all candidate keys.

$ (AB)^+ = {A, B, C, D, E, G} = R \ (A)^+ = {A} \not= R \ (B)^+ = {B, C, D, E, G} \not= R \ $

Therefore, AB is a *candidate key* (minimal superkey).

$ (ADE)^+ = {A, D, E, G} \not= R \ (G)^+ = {D, E, G} \not= R \ $

I think AB is the only *candidate key*.

> d. The final decomposition.

$$A, D, E, G, B, C, G, A, B$$

> b. A canonical cover for F, along with an explanation of the steps you took to generate it.

We use the algorithm given on Figure 7.9.

Set $F_c = F = AB \rightarrow CD, ADE \rightarrow GDE, B \rightarrow GC, G \rightarrow DE$.

Use the **union rule** to replace any dependencies in F_c of the form $\alpha_1 \rightarrow \beta_1$ and $\alpha_1 \rightarrow \beta_2$ with $\alpha_1 \rightarrow \beta_1\beta_2$.

But looking at the functional dependencies given in F_c we see that there are no such dependencies.

Now, we find a functional dependency $\alpha \rightarrow \beta$ in F_c with an extraneous attribute either in α or in β. If an extraneous attribute is found, we delete it from $\alpha \rightarrow \beta$ in F_c. We continue this iteration until F_c does not change.

Looking at the functional dependency $AB \rightarrow CD$ we see that C is extraneous. Since the functional dependency $B \rightarrow C$ can be inferred from the functional dependency $(B \rightarrow GC) \in F_c$ by the **Decomposition rule**.

So the current state of F_c looks like:-

$$F_c = AB \rightarrow D, ADE \rightarrow GDE, B \rightarrow GC, G \rightarrow DE$$

We see that E on the right hand side of the functional dependency $ADE \rightarrow GDE$ is extraneous. So let's delete that one. Then F_c becomes:-

$$F_c = AB \rightarrow D, ADE \rightarrow GD, B \rightarrow GC, G \rightarrow DE$$

Attribute D on the right hand side of the functional dependency $ADE \rightarrow GD$ is also extraneous. So F_c becomes:-

$$F_c = AB \rightarrow D, ADE \rightarrow G, B \rightarrow GC, G \rightarrow DE$$

By using the functional dependencies $B \rightarrow GC, G \rightarrow DE$ and **Decomposition rule** followed by **Transitivity rule**, it can be shown that $B \rightarrow D$ holds. Thus we can delete the functional dependency $AB \rightarrow D$.

$$F_c = ADE \rightarrow G, B \rightarrow GC, G \rightarrow DE$$

Since we can not find any other extraneous attributes, we claim that F_c is a canonical cover for F.

> c. The remaining steps of the algorithm, with explanation.

Define $F_c := ADE \rightarrow G, B \rightarrow GC, G \rightarrow DE$

Define
$R_1 := A, D, E, G$
$R_2 := B, C, G$
$R_3 := D, E, G$

We see that none of the schemas $R_i :: \forall i, 1 \leq i \leq 3$ contains a candidate key. Thus we define R_4 as:

$R_4 := A, B$

Then we delete R_3 since $R_3 \subseteq R_1$.

Thus the follwing is a 3NF decomposition of our schema R.

$$R_1, R_2, R_4 = A, D, E, G, B, C, G, A, B$$

7.34 Consider the schema $R = (A, B, C, D, E, G, H)$ and the set F of functional dependencies:

$$AB \rightarrow CD$$
$$D \rightarrow C$$
$$DE \rightarrow B$$
$$DEH \rightarrow AB$$
$$AC \rightarrow DC$$

Use the 3NF decomposition algorithm to generate a 3NF decomposition of R, and show your work. This means:

a. A list of all candidate keys

b. A canonical cover for F

c. The steps of the algorithm, with explanation

d. The final decomposition

> a. A list of all candidate keys

Candidate keys that I can come up with:-

$$A, B, E, G, H, D, E, G, H$$

> b. A canonical cover for F

Set $$ F_c := F = {AB \rightarrow CD, D \rightarrow C, DE \rightarrow B, DEH \rightarrow AB, AC \rightarrow DC} $$

C is extraneous in the right hand side of $AC \rightarrow DC$

$$F_c = AB \rightarrow CD, D \rightarrow C, DE \rightarrow B, DEH \rightarrow AB, AC \rightarrow D$$

C is extraneous in the right hand side of $AB \rightarrow CD$. This is because $AB \rightarrow C$ can be inferred from $AB \rightarrow D$, $D \rightarrow C$ and **Transitivity rule**.

$$F_c = AB \rightarrow D, D \rightarrow C, DE \rightarrow B, DEH \rightarrow AB, AC \rightarrow D$$

B is extraneous in $DEH \rightarrow AB$. This is because $DEH \rightarrow B$ can be inferred from $DE \rightarrow B$ and applying **Augmentation rule** followed by **Decomposition rule**.

$$F_c = AB \rightarrow D, D \rightarrow C, DE \rightarrow B, DEH \rightarrow A, AC \rightarrow D$$

Since I cannot find any more extraneous attributes, the canonical cover of F is:-

$$F_c = AB \rightarrow D, D \rightarrow C, DE \rightarrow B, DEH \rightarrow A, AC \rightarrow D$$

> c. The steps of the algorithm, with explanation

Define $F_c := AB \rightarrow D, D \rightarrow C, DE \rightarrow B, DEH \rightarrow A, AC \rightarrow D$

Define
$R_1 := A, B, D$
$R_2 := C, D$
$R_3 := B, D, E$
$R_4 := A, D, E, H$
$R_5 := A, C, D$

We see that none of the schemas $R_i :: \forall i, 1 \leq i \leq 5$ contains a candidate key. Thus we define R_6 as:

$R_6 := D, E, G, H$

Then we delete R_2 since $R_2 \subseteq R_5$.

Thus our result is

$$R_1, R_3, R_4, R_5, R_6 = A, B, D, B, D, E, A, D, E, H, A, C, D, D, E, G, H$$

> d. The final decomposition

$$A, B, D, B, D, E, A, D, E, H, A, C, D, D, E, G, H$$